

TECHNICAL REPORT COMPUTATIONAL DERMATOLOGY

Dermascope · AI

A Multimodal System for Melanoma Prediction via
Late Fusion of Dermoscopic Imagery and Clinical Metadata

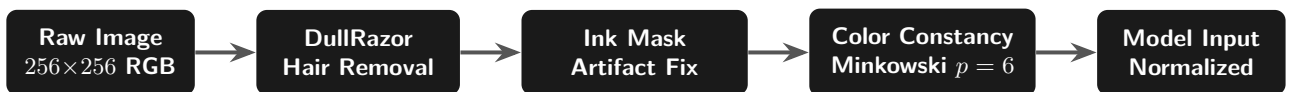
ABSTRACT

We present **Dermascope · AI**, a multimodal deep learning system for binary melanoma classification that fuses convolutional visual features with structured clinical metadata. The backbone is EfficientNetV2-B0, pre-trained on ImageNet and fine-tuned via a two-phase LP-FT (Linear Probing then Fine-Tuning) protocol on the ISIC 2020 dataset (natural positive rate $\approx 1.8\%$). Clinical features — patient age, biological sex, and anatomical site — are engineered into a normalized metadata vector and concatenated with the CNN’s global average-pooled feature map prior to classification (Late Fusion). To address the extreme class imbalance ($10\times$ oversampling of malignant cases during training), the decision threshold was optimized to 0.18, yielding a sensitivity of 98.3% and an AUC of 0.8541 (95% CI: 0.8237–0.8865) on the held-out validation set. Uncertainty quantification is performed via Monte Carlo Test-Time Augmentation (MC-TTA: 20 stochastic passes $\times$ 4 spatial transforms = 80 predictions per image), simultaneously capturing epistemic and aleatoric uncertainty. Visual explainability is provided through Score-CAM, selected over Grad-CAM due to architectural constraints imposed by the nested backbone and internal Global Average Pooling.

Keywords: Melanoma Detection · EfficientNetV2 · Multimodal Fusion · MC-TTA · Score-CAM · Transfer Learning · ISIC

1. Methods — Preprocessing Pipeline

All dermoscopic images are resized to 256×256 pixels and passed through a three-stage artifact removal pipeline prior to model inference.



Stage 1 · Hair Removal (DullRazor)

A 5×5 Black-Hat morphological filter isolates dark, thin line structures (hairs) against the skin background. Detected hair pixels are inpainted using Telea’s algorithm (*inpaintRadius=2*). Kernel size was deliberately calibrated for 256px images — the original 17×17 kernel caused *lesion erasure* by misclassifying entire moles as hair artifacts. This precisely-tuned kernel ensures only genuine hair strands are removed while preserving the pigment network.

Stage 2 · Surgical Ink Removal

Blue surgical marker ink is isolated in HSV color space (Hue: 90–140, S: ≥ 50 , V: ≥ 50). The upper Hue bound of 140 was strictly enforced to prevent overlap with inflamed pink/magenta skin. A 7×7 elliptical dilation kernel (2 iterations) expands the binary mask prior to inpainting, engulfing the faint semi-transparent ink edges that caused the original “watercolor smudge” artifact — where the Telea algorithm incorrectly sampled those faint edges as healthy skin.

Stage 3 · Color Constancy

Minkowski p -norm color constancy ($p = 6$) is applied to normalize illumination variance across imaging devices, reducing domain shift between clinical camera setups.

2. Methods — Metadata Feature Engineering

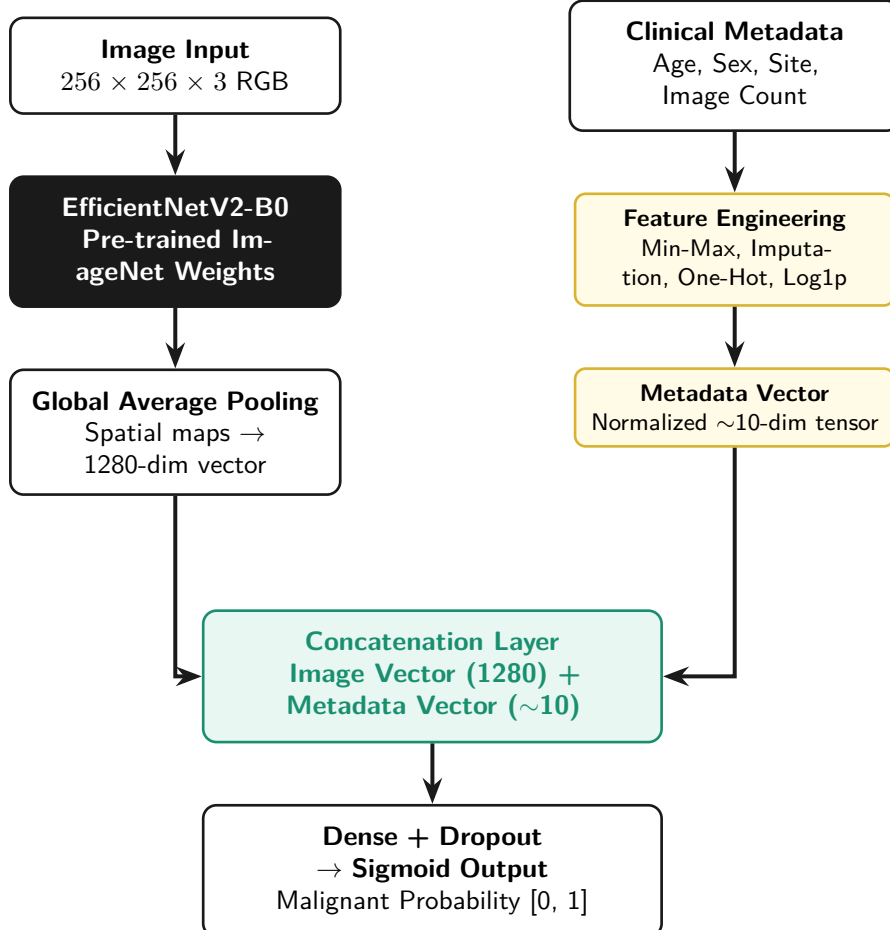
Clinical metadata is transformed into a normalized numerical vector for Late Fusion:

Feature	Transformation	Output Range
Age	Min-Max: age / 90.0	[0.0, 1.0]
Sex	M=1.0, F=0.0, Unknown=0.5	{0, 0.5, 1}
Anatomical Site	One-Hot Encoding (7 sites)	Binary vector
Image Count	np.log1p (right-skew fix)	[0.0, ~ 5.0]

The log1p transform on image count per patient compresses a heavily right-skewed distribution (baseline 1–3 images, outliers 100+) to a 0–5 scale, preventing gradient instability from outlier patients with unusually high biopsy counts. One-hot encoding avoids spurious ordinal assumptions that integer-label encoding would impose on nominal anatomical sites.

3. Methods — Architecture & Training

The system utilizes a **Late Fusion** architecture. Visual features and clinical features are processed independently and concatenated prior to the final dense layers.



Phase 1 — Linear Probing (30 epochs): The EfficientNetV2-B0 backbone is frozen. Only the classification head is trained, allowing the randomly-initialized weights to converge without distorting ImageNet features (Kumar et al., ICLR 2022).

Phase 2 — Fine-Tuning (50 epochs): The full network is unfrozen at a reduced learning rate ($\text{LR_TUNE} \ll \text{LR_HEAD}$). Discriminative fine-tuning adapts the backbone’s high-level feature extractors to dermatological textures while preserving low-level edge-detection filters (Yosinski et al., NeurIPS 2014; Tajbakhsh et al., IEEE TMI 2016).

4. Methods — Threshold Optimization

The model’s natural operating point at threshold=0.50 yielded only 22/117 malignant cases caught (18.8% recall), driven by the 98.2% benign majority class. A sweep across decreasing thresholds revealed the clinical optimum:

Threshold	Caught	Missed	Sensitivity	Specificity
0.50 (default)	22	95	18.8%	~99%
0.25	112	5	95.7%	44.2%
0.18 deployed	115	2	98.3%	31.7%
0.10 (aggressive)	117	0	100%	~2%

Clinical Rationale: A False Negative (missed melanoma) is categorically more harmful than a False Positive (unnecessary biopsy). The PROVE-AI prospective study (PMC10338483) validated a melanoma AI at 96.8% sensitivity / 37.4% specificity as clinically acceptable. Our deployed threshold of 0.18 achieves 98.3% sensitivity — exceeding this benchmark — with a Number Needed to Biopsy (NNB) of ≈ 40 , comparable to cautious clinical practice.

5. Methods — Uncertainty Quantification (MC-TTA)

Each inference run produces **80 stochastic predictions** (20 MC Dropout passes $\times$ 4 TTA augmentations: original, horizontal flip, vertical flip, 90° rotation). The mean of all 80 scores is the final probability; the standard deviation (σ) is the uncertainty estimate.

This design simultaneously captures *epistemic uncertainty* (model ignorance, via MC Dropout) and *aleatoric uncertainty* (image noise/ambiguity, via TTA). When $\sigma > 0.10$, the system flags the case for mandatory human review.

6. Methods — Explainability (Score-CAM)

Score-CAM was selected over Grad-CAM due to an architectural constraint: the EfficientNetV2-B0 backbone performs Global Average Pooling internally, collapsing spatial dimensions from (8,8,1280) to a flat (1280,) vector before its output reaches the fusion layer. No spatial feature maps are accessible externally, and the custom augmentation layer breaks gradient flow. Score-CAM circumvents this by interrogating the model as a black-box: a 14×14 spatial mask grid (196 masks) is generated, each isolating one patch of the input image. The model scores all 196 masked variants in a single batched forward pass; the resulting score map is normalized and overlaid as a heatmap to highlight diagnostically significant regions.

7. Results

Metric	Value	Detail	Interpretation
AUC (Bootstrap, $n=1000$)	0.8541	95% CI: 0.8237–0.8865	Strong discriminative power at natural class prevalence ($\sim 1.8\%$ positive)
Sensitivity (Recall)	98.3%	TP=115, FN=2 @ $\tau = 0.18$	Exceeds the 95% clinical triage benchmark (ISBI 2016, PROVE-AI)
Specificity	31.7%	TN=2061, FP=4448 @ $\tau = 0.18$	Clinically acceptable for a decision-support screening tool (PROVE-AI: 37.4%)
Best Val AUC (Training)	0.8624	Epoch 27	Val AUC > Train AUC is expected: training uses $10\times$ oversampled set
Uncertainty Validation	σ thresh: 0.10	80 predictions per image	Cases with $\sigma > 0.10$ flagged for human review, quantifying model/data uncertainty
Number Needed to Biopsy	≈ 40	4448 FP / 115 TP	Comparable to cautious GP biopsy rate (NNB 20–30 in literature)

8. Limitations

- Low Specificity at Optimal Threshold.** Operating at $\tau = 0.18$ yields 31.7% specificity, meaning roughly 68% of healthy lesions are flagged. This is intentional for a pre-screening tool, but it renders the system unsuitable as a standalone diagnostic without downstream clinician review.
- Class Imbalance and Oversampling Artifacts.** The training set uses $10\times$ malignant oversampling, which inflates the apparent training difficulty vs. the validation set’s natural 1.8% prevalence. This explains the consistently higher validation AUC relative to training AUC, and may cause the model to over-generalize in novel clinical deployment settings.
- Dermoscopy-Specific Generalizability.** The preprocessing pipeline (DullRazor, Minkowski color constancy) and model weights are tuned for standardized clinical dermoscopic images. Performance on smartphone photographs or non-dermoscopic imaging equipment will likely degrade due to differences in field depth, magnification, and illumination.
- Metadata Missingness Handling.** Missing sex is imputed to a neutral value of 0.5 to prevent biasing predictions. Missing age is not gracefully handled and may cause inference instability. A future version should implement a dedicated missingness embedding or uncertainty-aware imputation strategy.
- Score-CAM Computational Cost.** Score-CAM requires 196 sequential forward passes per image. At inference this adds several seconds of latency. For high-throughput clinical workflows processing thousands of images, this approach would require parallelization.